

Lecture 10

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Convention. The ground set is N , and $M = (N, \mathcal{I})$ is a matroid of rank r with set of bases $\mathcal{B}(M)$. We assume $r \geq 1$; the rank-zero case is trivial. Throughout, $f : 2^N \rightarrow \mathbb{R}_{\geq 0}$ is normalized, monotone, and submodular: $f(\emptyset) = 0$. For $i \notin S$, write $f(i \mid S) = f(S + i) - f(S)$. Its multilinear extension is

$$F(x) = \mathbb{E}[f(R(x))],$$

where $R(x)$ contains each i independently with probability x_i . The notation $x \vee y$ denotes the coordinatewise maximum. By monotonicity, an optimal independent set can be extended to a base, so we compare all algorithms with a fixed optimal base O .

1 Why Ordinary Local Search Stops at One Half

The one-exchange neighborhood of a base A consists of all bases $A - a + b$, where $a \in A$ and $b \notin A$. Ordinary local search optimizes the objective f itself over this neighborhood. Its factor $1/2$ is tight even for a coverage function under a partition matroid.

Example 1.1 (A tight one-exchange local optimum). Let the coverage universe be $\{x, y\}$, and partition four items into $P_1 = \{a_1, a_2\}$ and $P_2 = \{b_1, b_2\}$. A base selects one item from each part. The sets covered by the items are

$$C(a_1) = \{x\}, \quad C(a_2) = \{y\}, \quad C(b_1) = \emptyset, \quad C(b_2) = \{x\}.$$

For $f(S) = |\bigcup_{i \in S} C(i)|$, the base $A = \{a_1, b_1\}$ has value 1. Replacing either selected item by the other item in its part still gives value 1, so A is a one-exchange local optimum. However, $O = \{a_2, b_2\}$ has value 2. Thus $f(A)/f(O) = 1/2$.

The obstruction is not the neighborhood alone. We can keep the same one-element exchanges but optimize a more informative potential. This is the idea of *non-oblivious* local search.

2 The Filmus–Ward Potential

For $S \subseteq N$, define

$$\tilde{f}(S) = \int_0^1 F(p\mathbf{1}_S) \frac{e^p}{p} dp. \tag{1}$$

Filmus and Ward use the same potential scaled by $1/(e - 1)$; the positive constant has no effect on local optima [3].

The normalization $f(\emptyset) = 0$ is essential in (1). Indeed,

$$F(p\mathbf{1}_S) = p \sum_{i \in S} f(\{i\}) + O(p^2) \quad (p \downarrow 0),$$

so the apparent singularity at $p = 0$ is removable. For fixed p , if Q_p includes every element independently with probability p , then

$$F(p\mathbf{1}_S) = \mathbb{E}[f(S \cap Q_p)].$$

For every realization of Q_p , the map $S \mapsto f(S \cap Q_p)$ is monotone and submodular. Consequently, $\tilde{f}$ is also monotone and submodular.

Definition 2.1 (Non-oblivious local optimum). A base A is a one-exchange local optimum of $\tilde{f}$ if

$$\tilde{f}(A) \geq \tilde{f}(A - a + b)$$

for every $a \in A$ and $b \notin A$ for which $A - a + b$ is a base.

Algorithm 1 Idealized non-oblivious local search

Input: A matroid $M = (N, \mathcal{I})$ and a normalized monotone submodular function f .

Choose an arbitrary base $A \in \mathcal{B}(M)$.

while there are $a \in A$ and $b \notin A$ such that $A - a + b \in \mathcal{B}(M)$ and

$\tilde{f}(A - a + b) > \tilde{f}(A)$ **do**

$A \leftarrow A - a + b$.

end while

return A .

Algorithm 1 is written to expose the locality argument. Exact evaluation of the integral and unrestricted strict-improvement steps do not by themselves give a polynomial-time value-oracle algorithm; we return to this issue after the analysis.

2.1 Two ingredients

The first ingredient is the bijective basis-exchange property.

Lemma 2.2 (Brualdi's exchange bijection [1]). *For bases $A, O \in \mathcal{B}(M)$, there is a bijection $h : A \rightarrow O$ such that $h(a) = a$ for $a \in A \cap O$ and*

$$A - a + h(a) \in \mathcal{B}(M) \quad \text{for every } a \in A.$$

The second ingredient converts a sum of multilinear marginals into a global improvement.

Lemma 2.3 (Gradient domination). *For every $x \in [0, 1]^N$ and $B \subseteq N$,*

$$\sum_{i \in B} \partial_i F(x) \geq F(x \vee \mathbf{1}_B) - F(x). \tag{2}$$

Proof. Let $d = (x \vee \mathbf{1}_B) - x$. Then $0 \leq d \leq \mathbf{1}_B$. Submodularity implies that F is concave along every nonnegative direction, and therefore

$$F(x + d) - F(x) \leq \langle \nabla F(x), d \rangle.$$

Monotonicity gives $\nabla F(x) \geq 0$. Hence $\langle \nabla F(x), d \rangle \leq \langle \nabla F(x), \mathbf{1}_B \rangle$, proving the claim. $\square$

2.2 Analysis of a local optimum

Lemma 2.4 (The local-optimality inequality). *Let A be a one-exchange local optimum of $\tilde{f}$, let O be a base, and let $h : A \rightarrow O$ be the map in Lemma 2.2. Then*

$$\sum_{a \in A \setminus O} \tilde{f}(a \mid A - a) \geq \sum_{b \in O \setminus A} \tilde{f}(b \mid A). \quad (3)$$

Proof. For every $a \in A \setminus O$, local optimality and submodularity of $\tilde{f}$ give

$$\begin{aligned} \tilde{f}(a \mid A - a) &= \tilde{f}(A) - \tilde{f}(A - a) \\ &\geq \tilde{f}(A - a + h(a)) - \tilde{f}(A - a) \\ &= \tilde{f}(h(a) \mid A - a) \geq \tilde{f}(h(a) \mid A). \end{aligned}$$

Sum over $a \in A \setminus O$ and use that h maps $A \setminus O$ bijectively onto $O \setminus A$. $\square$

Theorem 2.5 (Non-oblivious locality theorem [3]). *If A is a one-exchange local optimum of $\tilde{f}$ and O is an optimal base, then*

$$f(A) \geq \left(1 - \frac{1}{e}\right) f(O).$$

Proof. Add the deletion marginals indexed by $A \cap O$ to both sides of (3). This gives the correctly oriented inequality

$$\sum_{a \in A} \tilde{f}(a \mid A - a) \geq \sum_{b \in O \setminus A} \tilde{f}(b \mid A) + \sum_{a \in A \cap O} \tilde{f}(a \mid A - a). \quad (4)$$

Multilinearity makes $\partial_i F$ independent of coordinate x_i . Thus, for $b \notin A$ and $a \in A$,

$$\tilde{f}(b \mid A) = \int_0^1 \partial_b F(p \mathbf{1}_A) e^p dp, \quad (5)$$

$$\tilde{f}(a \mid A - a) = \int_0^1 \partial_a F(p \mathbf{1}_A) e^p dp. \quad (6)$$

The right-hand side of (4) is therefore

$$\int_0^1 e^p \sum_{o \in O} \partial_o F(p \mathbf{1}_A) dp.$$

By Lemma 2.3 and monotonicity,

$$\begin{aligned} \sum_{o \in O} \partial_o F(p \mathbf{1}_A) &\geq F(p \mathbf{1}_A \vee \mathbf{1}_O) - F(p \mathbf{1}_A) \\ &\geq f(O) - F(p \mathbf{1}_A). \end{aligned}$$

Set $q(p) = F(p \mathbf{1}_A)$. The chain rule gives $q'(p) = \sum_{a \in A} \partial_a F(p \mathbf{1}_A)$. Substituting (5)–(6) into (4) now yields

$$\int_0^1 e^p q'(p) dp \geq \int_0^1 e^p (f(O) - q(p)) dp.$$

Equivalently,

$$\int_0^1 e^p (q'(p) + q(p)) dp \geq (e - 1) f(O).$$

The integrand on the left is the derivative of $e^p q(p)$, so

$$eq(1) - q(0) \geq (e - 1)f(O).$$

Finally, $q(0) = f(\emptyset) = 0$ and $q(1) = f(A)$. □

Remark (From the structural theorem to an algorithm). Theorem 2.5 explains the potential, but the idealized Algorithm 1 is not yet a runtime analysis. Filmus and Ward estimate the potential by sampling and accept only sufficiently large improvements. This gives a randomized polynomial-time $1 - 1/e - \eta$ approximation for every fixed $\eta > 0$; partial enumeration recovers the tight $1 - 1/e$ guarantee [3].

3 Continuous Greedy, Rounded on the Fly

The continuous-greedy trajectory satisfies

$$x'(t) = \mathbf{1}_{Z(t)}, \quad Z(t) \in \arg \max_{B \in \mathcal{B}(M)} \sum_{i \in B} \partial_i F(x(t)), \quad (7)$$

where linear optimization over a matroid is performed by the greedy algorithm [2]. For $t > 0$, put $y(t) = x(t)/t$. Then

$$y'(t) = \frac{\mathbf{1}_{Z(t)} - y(t)}{t}. \quad (8)$$

Equivalently, over a short time step δ ,

$$y(t + \delta) = \frac{t}{t + \delta} y(t) + \frac{\delta}{t + \delta} \mathbf{1}_{Z(t)} + o(\delta).$$

This suggests maintaining an actual random base whose conditional infinitesimal drift mirrors the scaled trajectory, and rounding each infinitesimal update immediately.

Suppose the current base is A and Z is the target base. Choose a Brualdi map $h : A \rightarrow Z$, then choose a uniformly from A and replace it by $h(a)$. Conditionally on A ,

$$\mathbb{E}[\mathbf{1}_{A-a+h(a)} - \mathbf{1}_A] = \frac{1}{r}(\mathbf{1}_Z - \mathbf{1}_A).$$

If these swaps occur at rate r/t , their expected drift is exactly $(\mathbf{1}_Z - \mathbf{1}_A)/t$, matching (8). This is the Greedy Swap Poisson Process of Ganz Rozenman, Kulik, Schwartz, and Singh [4].

4 The Greedy Swap Poisson Process

The rate r/t is singular at $t = 0$. A finite algorithm must therefore start at a parameter $\varepsilon > 0$. The nonhomogeneous Poisson process on $[\varepsilon, 1]$ has intensity

$$\lambda(t) = \frac{r}{t}.$$

Starting from time t , its next event time τ satisfies

$$\Pr(\tau > s) = \exp\left(-\int_t^s \frac{r}{u} du\right) = \left(\frac{t}{s}\right)^r.$$

Thus, if U is uniform on $(0, 1)$, we may sample $\tau = tU^{-1/r}$.

Algorithm 2 Greedy Swap Poisson Process

Input: $M = (N, \mathcal{I})$ of rank r , f , and $\varepsilon \in (0, 1)$.

Choose an arbitrary base $A \in \mathcal{B}(M)$ and set $t \leftarrow \varepsilon$.

while $t < 1$ **do**

 Draw U uniformly from $(0, 1)$ and set $\tau \leftarrow tU^{-1/r}$.

if $\tau > 1$ **then return** A .

$t \leftarrow \tau$.

 Choose $Z \in \arg \max_{B \in \mathcal{B}(M)} \sum_{i \in B} \partial_i F(t\mathbf{1}_A)$.

 Find a bijection $h : A \rightarrow Z$ as in Lemma 2.2.

 Choose a uniformly from A and set $A \leftarrow A - a + h(a)$.

end while

return A .

If $a \in A \cap Z$, the map fixes a and the update is a harmless no-op. The number of events is Poisson with mean

$$\int_{\varepsilon}^1 \frac{r}{t} dt = r \log \frac{1}{\varepsilon}. \quad (9)$$

Theorem 4.1 (Ganz Rozenman–Kulik–Schwartz–Singh [4]). *Let $A(1)$ be the base returned by Algorithm 2. Then*

$$\mathbb{E}[f(A(1))] \geq (1 - e^{-(1-\varepsilon)}) f(O) \geq (1 - \varepsilon) \left(1 - \frac{1}{e}\right) f(O). \quad (10)$$

The expected number of exchange events is $r \log(1/\varepsilon)$.

4.1 Infinitesimal analysis

Let $A(t)$ denote the random current base and define the deterministic quantity

$$Q(t) = \mathbb{E}[F(t\mathbf{1}_{A(t)})]. \quad (11)$$

It is important to take the expectation: the sample path $F(t\mathbf{1}_{A(t)})$ jumps at Poisson event times.

For a short interval $(t, t + \delta]$, let X be the number of events. Its mean is

$$\Lambda(t, \delta) = \int_t^{t+\delta} \frac{r}{s} ds = \frac{r}{t} \delta + O(\delta^2).$$

Consequently,

$$\Pr(X = 0) = 1 - \frac{r}{t} \delta + O(\delta^2),$$

$$\Pr(X = 1) = \frac{r}{t} \delta + O(\delta^2),$$

$$\Pr(X \geq 2) = O(\delta^2).$$

The quadratic remainder in the last line is what makes a first-order generator calculation valid.

Condition on $A(t) = A$, and abbreviate $w_i = \partial_i F(t\mathbf{1}_A)$. If no event occurs, increasing t contributes the radial derivative

$$\frac{d}{dt}F(t\mathbf{1}_A) = \sum_{a \in A} w_a. \quad (12)$$

At an event, let Z and $h : A \rightarrow Z$ be chosen as in the algorithm. For the update indexed by $a \in A$, multilinearity gives

$$\begin{aligned} & F(t\mathbf{1}_{A-a+h(a)}) - F(t\mathbf{1}_A) \\ &= (F(t\mathbf{1}_{A-a+h(a)}) - F(t\mathbf{1}_{A-a})) - (F(t\mathbf{1}_A) - F(t\mathbf{1}_{A-a})) \\ &= t \partial_{h(a)} F(t\mathbf{1}_{A-a}) - t \partial_a F(t\mathbf{1}_A) \\ &\geq t(w_{h(a)} - w_a). \end{aligned} \quad (13)$$

The final inequality is diminishing returns, since $t\mathbf{1}_{A-a} \leq t\mathbf{1}_A$.

Events occur at rate r/t , and each $a \in A$ is selected with probability $1/r$. Combining the radial contribution (12) with the average jump contribution from (13), the terms indexed by A cancel. More precisely, if $D^+(t, A)$ denotes the conditional right drift given $A(t) = A$, then

$$\begin{aligned} D^+(t, A) &\geq \sum_{a \in A} w_a + \frac{1}{t} \sum_{a \in A} t(w_{h(a)} - w_a) \\ &= \sum_{a \in A} w_{h(a)} = \sum_{z \in Z} w_z. \end{aligned} \quad (14)$$

Because Z is a maximum-gradient base,

$$\sum_{z \in Z} w_z \geq \sum_{o \in O} w_o.$$

Lemma 2.3 and monotonicity then imply

$$\begin{aligned} \sum_{o \in O} w_o &\geq F(t\mathbf{1}_A \vee \mathbf{1}_O) - F(t\mathbf{1}_A) \\ &\geq f(O) - F(t\mathbf{1}_A). \end{aligned}$$

Average (14) over $A(t)$. We obtain the differential inequality

$$Q'_+(t) \geq f(O) - Q(t). \quad (15)$$

Multiplying by e^t and integrating from ε to 1 gives

$$\begin{aligned} Q(1) &\geq e^{-(1-\varepsilon)} Q(\varepsilon) + (1 - e^{-(1-\varepsilon)}) f(O) \\ &\geq (1 - e^{-(1-\varepsilon)}) f(O), \end{aligned}$$

where the last inequality uses nonnegativity. At time 1, $F(\mathbf{1}_{A(1)}) = f(A(1))$. Finally,

$$1 - e^{-(1-\varepsilon)} \geq (1 - \varepsilon)(1 - e^{-1}),$$

which proves Theorem 4.1.

Remark (Why the process cannot begin at zero). Formally sending $\varepsilon \downarrow 0$ in (10) recovers $1 - 1/e$. However, (9) then diverges. Thus an algorithm with a finite expected number of swaps begins at positive ε and has the guarantee stated in Theorem 4.1; an exact $1 - 1/e$ claim from a process started at $t = 0$ would be incorrect.

Summary. Ordinary one-swap local search can be stuck at factor $1/2$. The Filmus–Ward potential reweights multilinear values across all scales $p \in [0, 1]$; exact local optimality for this potential certifies the tight $1 - 1/e$ bound. The Greedy Swap Poisson Process instead rounds continuous greedy on the fly. Its rate r/t makes the loss from removing the current element cancel the radial derivative, leaving exactly the maximum-gradient gain. Starting at $\varepsilon > 0$ gives factor $1 - e^{-(1-\varepsilon)}$ using $r \log(1/\varepsilon)$ expected exchanges.

References

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